Martin Drieux

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Education

Carnegie Mellon University (CMU) Pittsburgh, PA
Master of Science in Machine Learning Expected Dec. 2026
Relevant Coursework: Deep Learning Systems, Deep Reinforcement Learning, Probabilistic Graphical Models.

École Polytechnique – France's top university for science and engineering Paris, France
Cycle Ingénieur (B.S./M.S. equivalent, 5-year program), Applied Math, Physics and ML. **CGPA: 3.96/4.0.** Jun. 2025
Relevant Coursework: Machine Learning for Scientific Computing, Control Theory, Statistics, Optimization.

Experience

Mistral AI, Research Scientist Intern – Post-training May 2026 – Aug. 2026, Palo Alto, CA
• **Test writing** (main project): analyzed the model's test-writing capabilities, designed **evals** and **synthetic data pipelines** to create training tasks, and improved the model with SFT and RL from **15% to 32%** on SWE Atlas – Test Writing.
• Contributed to improving the tests used as **verifiers** in synthetic environments for general coding capabilities.
• Environments and SFT data from both projects were added to the training of **Mistral Large 4**.

CMU – Research Assistant, Generative Modeling Sep. 2025 – Present, Pittsburgh, PA
• **Equilibrium Flow Maps** (co-first author, Prof. Nicholas Boffi): flow maps for time-free (autonomous) generative dynamics: one model does one-step generation or iterative refinement.
• ImageNet-256: **FID 2.83 in 1 call, 1.67 in 8**; unlike time-dependent flow maps, it improves steadily with more steps.
• **Mean Flow Improvements** (Prof. Zico Kolter's group, with Zhengyang Geng, Sep. 2025 – Apr. 2026): stochastic Mean Flow sampling, likelihood computation, and training algorithms based on **fixed-point methods**.

BAIR – UC Berkeley, Visiting Student Researcher (Prof. Alexandre Bayen) Mar. 2025 – Aug. 2025, Berkeley, CA
• Designed **NFVM**, an ML-based PDE solver with convergence guarantees, achieving an **8× accuracy improvement** over classical numerical solvers at equal compute.
• Extended NFVM to Hamilton–Jacobi equations for dynamic optimization and control.

Avencore, Consultant Intern Jun. 2024 – Sep. 2024, Paris, France
• **$100M+ cost-reduction** project (uranium extraction): built linear programming and facility location models.

Gendarmerie Nationale, Deputy Company Commander Sep. 2022 – Mar. 2023, Toulouse, France
• Developed forensic computing tools for victim identification on seized devices; led training sessions for 15 personnel.

Publications

Equilibrium Flow Maps. J. Lin*, **M. Drieux***, J. Huang, S. Dandapanthula, N. M. Boffi. Under review at ICLR 2027. [PDF]

Supervised and Unsupervised Neural Network Solver for First Order Hyperbolic Nonlinear PDEs. **Co-first author**, under A. M. Bayen et al. arXiv 2026.

Awards & Honors

International Physicists' Tournament (2024): year-long open physics research problems (6-person team). **Silver Medal** internationally (**2nd of 21 countries**), **Gold Medal** nationally; best scientific project **(1/112)** at Polytechnique.

Selected Projects

Deep Learning Systems (CMU) Aug. 2025 – Dec. 2025
• Reimplemented the *Needle* deep learning framework (PyTorch-style autograd) from scratch, then used it to build a fixed-point (Broyden) solver with implicit differentiation and custom **CUDA kernels**.

Attention Flow/Rollout – Interpretability for Transformers Jan. 2025 – Apr. 2025
• Implemented attention flow and attention rollout to explain transformer predictions; designed metrics that evaluate relevance by the drop in classification score.

NLP Kaggle Challenge (École Polytechnique) Sep. 2024 – Dec. 2024
• Built and optimized an event-classification pipeline on social media posts; ranked **1st of 49 teams**.

Skills

ML & programming: Python, PyTorch, NumPy, pandas, scikit-learn, C++, CUDA; LLM post-training (SFT, RL), Slurm.
Languages: French (native), English (fluent), Spanish (intermediate).